STUDYNEST | CLASS 11 PHYSICS

# MARKING SCHEME AND MODEL ANSWERS

## Chapter 1: Units and Measurements

**Maximum Marks: 40 | Suggested time: 60 minutes**

Use after attempting the paper. Accept equivalent correct methods.

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Seven.

kelvin (K).

Three.

[LT^-1].

[MLT^-2].

Yes.

No.

pascal (Pa).

12.5.

[MLT^-1].

ut and at^2 both have dimension L; homogeneous.

20 m s^-1.

Four.

Both are [ML^2T^-2].

Equating dimensions gives a=, b=-, so T proportional to sqrt(l/g).

4pir^3/3=33.5 cm^3.

[M^-1L^3T^-2].

51.68 cm^2 rounds to 52 cm^2.

a=, b=-; T proportional to sqrt(m/k).

18.2 cm^3; extra digits are unsupported.